Nikhil Mehta

Contact Information

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Overview

I am a Deep Learning Ph.D. student at Duke University. My research focus has primarily been on deep generative models and Bayesian neural networks. My recent published work was on interpretable latent space for graph neural network using structured prior, and implicit sampling by a (data) score function parameterized by a neural network. My ongoing research sits at the intersection of 3 things: Life-Long Learning, Implicit Parameter Inference for deep neural networks & Continual Learning for RL settings. I am also interested in applications of Bayesian Deep Learning in uncertainty estimation, ensemble learning and interpretability in neural networks.

Education and Research Experience

Duke University, Ph.D. (2018–Present)

- Computer Engineering: Deep Learning. GPA: 4.0
- Advisor: Dr. Lawrence Carin

Indian Institute of Technology (IIT) Kanpur, Project Associate (2017–2018)
Worked on the following research projects:

- Generative Model with Nonparametric prior for graphs. Presented at ICML 2019.
- Scan Registration for Autonomous Driving. Presented at ICRA 2018.
- Deep Topic Modeling for Multi-Label Learning. Presented at AISTATS 2019.

Delhi Technological University, B.Tech. (2010–2014)

- Degree: Software Engineering, First Class
- Department: Computer Science and Engineering
- Technical Head, Society of Software Engineering.

Publications

- **N. Mehta**, K. Liang & L. Carin, "Bayesian Nonparametric Weight Factorization for Continual Learning with Neural Networks" (Manuscript under review).

- A. Sarkar*, **N. Mehta*** & P. Rai, "Learning interpretable hierarchical node representations via Ladder Gamma VAE". Workshop on Graph Representation Learning, Neural Information Processing Systems (**NeurIPS**) 2019. Full version of this work has been accepted to appear in the proceedings of Association for Advancements of Artificial Intelligence (**AAAI**) 2020. [pdf]

- **N. Mehta**, L. Carin & P. Rai, "Stochastic Blockmodels meet Graph Neural Networks". International Conference on Machine Learning (**ICML**) 2019. [pdf]

- **N. Mehta***, Jiachang Liu*, Chenyang Tao & L. Carin, "Estimation and Sampling of Unnormalized Statistical Models with Stein Score Matching". Workshop on Stein's Method at International Conference on Machine Learning (**ICML**) 2019. [pdf] [poster]

- R. Panda, A. Pensia, **N. Mehta**, M.Zhou & P. Rai, "Deep Topic Models for Multi-label Learning". International Conference on Artificial Intelligence and Statistics (**AISTATS**), 2019. [pdf]

- **N. Mehta**, J. McBride & G. Pandey, "Robust and Fast 3D Scan Alignment using Mutual Information". International Conference on Robotics and Automation (**ICRA**) 2018. [pdf]

* Equal Contribution

Industry Experience

- **Software Engineer at Fitso** (Jan - Aug '16)
Was responsible for the android application development. Engineered the product that supported rapid growth from 20,000 users to 100,000 global users in less than six months. Application was selected as the Editors' Choice at the Google play store. [link]

- **Software Engineer at Zomato** (June '14 - Jan '16)
  - Was a core member of the technical team. Developed server side APIs for various consumer and enterprise products provided by Zomato across 22 countries including India, US and Canada. - Was part of the mobile development team. I was responsible for the development and release cycles of the primary Zomato application which had more than 5 million users. [link]

Open Source Projects & Internships

- **Zero-Shot Learning using GANs**
  Proposed a wasserstein generative-adversarial-network (W-GAN) to synthesize data for unseen classes conditioned on class attributes as an application in *Zero-Shot Learning*. [code]
- **Learning Disentangled Representations under Supervision**
  Implemented an extension of the vanilla recurrent variational auto-encoder. The architecture had a combination of encoder-generator and a discriminator (trained in wake-sleep fashion) that allowed the network to *learn structured disentangled representations* for text data. [code]
- **Learning Equivariant Representations**
  Implemented a capsule based architecture to *learn equivariant representations* which could be used for pose regression. Open-sourced a proof of concept implementation of "Transforming Autoencoders" to learn transformations. [code]
- **Research Intern at University of Birmingham, UK** (Dec '13 - Jan '14)
  Developed a paradigm and an experimental game in Python to investigate how people's perception of a hidden partner affect their behavior while competing against an artificial intelligent bot. Used the existing speech recognition model to create an end-end application for voice based commands.
- **Research Intern at University of Leeds, UK** (Jul - Aug '13)
  Designed a vision algorithm for cell (lamallepodia) tracking in DIC microscopic videos using entropy differences in subsequent frames. Real-time implementation was further improved by implementing the Extended Kalman filter.

Technical Skills

- **Deep Learning Frameworks**: Tensorflow, Pytorch.
- **Languages**: Python, CUDA for GPU, MATLAB, PHP, LaTeX.
- **Statistical Modeling**: Stan, Edward (Bayesian methods in deep networks).
- **Courses**: Machine Learning - Ph.D. Level (A+), Vector Space with Applications (A), Text Data Acquisition Analysis - NLP (A+), Bayesian Methods and Modern Applications (A), Deep Learning (A+), Applied Stochastic Processes (A), Learning in Adversarial Setting†, Reinforcement Learning at Scale†. †Ongoing

Teaching Experience

- **Project Mentor, Probabilistic Machine Learning, IIT Kanpur**
  Course Instructor: Dr. Piyush Rai
- **Technical Head, Society of Software Engineering, DTU**

Review Contributions

- International Conference on Machine Learning (ICML)
- International Conference on Learning Representations (ICLR)
- International Conference on Artificial Intelligence and Statistics (AISTATS)

Grants

- Duke ECE Department Fellowship worth $33,000 for the first year.
- Duke ECE Conference Travel Award for NeurIPS & ICML 2019.
- ICRA Travel Grant.